

Comparison of v1.4 and v1.3 System Monitor Codes

Both v1.4 and v1.3 are advanced Tkinter-based system monitor GUIs with similar core features: real-time resource graphs, process listing with sorting/search, theme toggle, and a floating widget mode. However, v1.4 introduces several improvements and optimizations over v1.3, making it the better version overall.

Key Improvements in v1.4

1. More Efficient Process List Updating

- **v1.4** updates only the changed rows in the process list (Treeview), using a dictionary (`self.process_items`) to track and update or insert only necessary rows. This avoids clearing and repopulating the entire list, reducing flicker and improving performance, especially with frequent updates^[1].
- **v1.3** clears and repopulates the entire process list on every update, which is less efficient and can cause UI flicker^[2].

2. More Accurate CPU Usage Display

- **v1.4** normalizes the CPU percent for each process by dividing by the number of logical CPUs, providing a more accurate per-core representation in multi-core systems^[1].
- **v1.3** displays the raw `cpu_percent` value, which can be misleading on multi-core systems^[2].

3. Improved Data Cleanup

- **v1.4** maintains a rolling window of up to 60 data points for all metrics, ensuring consistent graph history and memory usage^[1].
- **v1.3** only keeps 15 data points, which results in shorter graph histories and less smooth plots^[2].

4. Separate Network Speed Tracking for Widget and Main Window

- **v1.4** keeps separate network counters for the main window and the widget (`self.last_net_io_main` and `self.last_net_io_widget`), ensuring accurate and independent speed calculations in both views^[1].
- **v1.3** uses a single counter for both, which can cause inaccurate readings if both views are open simultaneously^[2].

5. Enhanced Theming and Style Handling

- **v1.4** centralizes and improves theme/style setup with a dedicated `setup_styles()` method, ensuring consistent appearance across widgets and easier theme switching^[1].
- **v1.3** sets styles in the UI setup but lacks a clear separation for theme logic, making future maintenance harder^[2].

6. Better Error Handling and Logging

- **v1.4** adds more robust error handling, especially for process details and termination, and logs errors more precisely. It also avoids unnecessary popups for non-critical errors (e.g., when a process no longer exists)^[1].
- **v1.3** logs errors but is less nuanced in distinguishing error types^[2].

7. Process Search and Highlighting

- Both versions support process search and highlight matches, but v1.4 integrates this with the improved process updating logic for smoother UI feedback^[1] ^[2].

Feature Comparison Table

Feature/Aspect	v1.4	v1.3
Process List Update	Incremental, efficient updates	Full clear and repopulate each time
CPU % Normalization	Per-core normalization	Raw percent
Data Retention (Graph)	60 points (smoother, longer history)	15 points (shorter history)
Network Speed Counters	Separate for widget/main	Shared counter (possible inaccuracies)
Theming/Styling	Centralized, modular, easier to extend	Mixed with UI setup
Error Handling	More robust, nuanced	Basic
Widget Mode	Present, with improved network stats	Present, but shared counters
Logging	Detailed, context-aware	Basic

Summary

v1.4 is the better version because it is more efficient, accurate, and maintainable. Its incremental process list updates, per-core CPU normalization, improved data retention, and separate network counters for widget/main window address key limitations in v1.3. The code structure is also more modular, making future enhancements easier.

v1.3 remains functional and feature-rich, but v1.4 is a clear upgrade in both user experience and code quality. ^[1] ^[2]

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- 1. paste.txt
- 2. paste-2.txt